

$v = \frac{dx}{dt}$ | $\omega = \frac{d\theta}{dt}$

$a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$ | $\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$

$x(t) = x(0) + v(0)t + \frac{1}{2}at^2$ | $\theta(t) = \theta(0) + \omega(0)t + \frac{1}{2}\alpha t^2$

$v(t)^2 - v(0)^2 = 2a(x(t) - x(0))$ | $\omega(t)^2 - \omega(0)^2 = 2\alpha(\theta(t) - \theta(0))$

$V_2^2 - V_1^2 = 2a(x_2 - x_1) = 2\frac{F}{m}\Delta x$

$\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = F\Delta x$

$\Delta k = k_2 - k_1 = F\Delta x$

kinetic energy work done by F



mass m | $I = m\|\|r\|\|^2$ moment of inertia

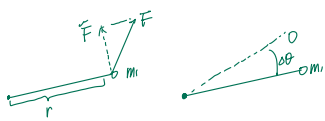
kinetic energy $k = \frac{1}{2}m\|\|v\|\|^2$ | $k = \frac{1}{2}I\|\|\omega\|\|^2$ kinetic energy

$k = \frac{1}{2}m\|\|v_{p/o}\|\|^2$

$= \frac{1}{2}m\|\|\omega \times \vec{r}_{p/o}\|\|^2$ $\omega \perp \vec{r}_{p/o}$

$= \frac{1}{2}m\|\|\vec{r}_{p/o}\|\|^2 \cdot \|\|\omega\|\|^2$

$= \frac{1}{2}I\|\|\omega\|\|^2$ $I = m\|\|r\|\|^2$ moment of inertia



Force F | $\tau = \vec{r} \times \vec{F}$ torque

torque: $\vec{r}_{p/o} \times \vec{F}$

work done: Fdx | $\tau d\theta$ work done

work done: $\|\|\vec{F}\|\| \cdot \|\|\vec{r}_{p/o}\|\| \cdot d\theta = \|\|\vec{r}_{p/o} \times \vec{F}\|\| d\theta$

apply force $F = ma$ | $\tau = I\alpha$ apply torque

work done = change in kinetic energy

$(\vec{r}_{p/o} \times \vec{F})d\theta = \frac{1}{2}I(\omega_2^2 - \omega_1^2)$

torque $\tau = \frac{1}{2}I 2\alpha d\theta$

$= I\alpha d\theta$

linear momentum $\vec{p} = m\vec{v}$

$\frac{d}{dt}\vec{p} = ma = F$

$\vec{h} = I\vec{\omega}$ angular momentum

$\frac{d}{dt}\vec{h} = I\vec{\alpha} = \vec{\tau}$

$\vec{h}_{p/o} = \vec{r}_{p/o} \times \vec{p}_{p/o} = m\vec{r}_{p/o} \times \vec{v}_{p/o}$

$\|\|\vec{h}\|\| = m\|\|r\|\| \cdot \|\|v\|\| = m\|\|r\|\|^2 \cdot \|\|\omega\|\| = I\|\|\omega\|\|$

$\vec{h}_{p/o} = I\omega$